DS603 · Probability for Data Science

Beyond the Bell Curve

Extreme Movements in Gold and Oil Returns

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Research Questions

1

Do daily gold and oil returns have heavier tails than a Normal distribution?

2

Do extreme movements in gold and oil occur jointly more often than would be expected under independence?

Data Sample and Series

32,932
raw Datastream rows

13,257
after removing 19,675 duplicates

6,501
dates with both prices

6,492
daily returns (8 gap-spanning dropped)

Source: LSEG Datastream via WRDS, daily closes, 4 Jan 2000 to 31 Dec 2025
Series: London gold bullion (USD/oz) and Brent crude M1 futures (USD/bbl)
Return: $r(t) = 100 \times \ln(P(t) / P(t-1))$, on dates when both markets have a price

Methodology Followed

Normal benchmark

The standard risk-model assumption: how many 3-SD days would it predict?

Q-Q plots

Show where, and in which direction, the data leave the bell curve.

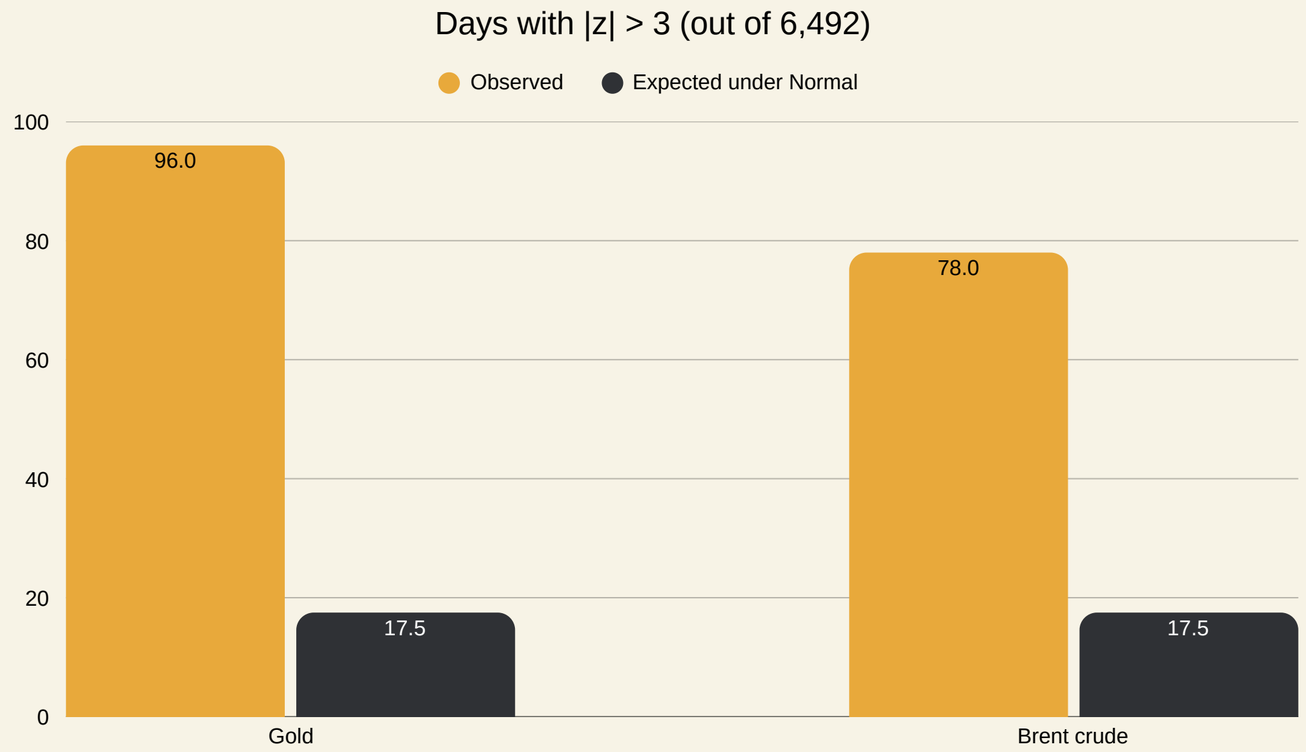
Chebyshev bound

A guarantee that needs no distribution: a sanity check on the tails.

Joint extremes + Fisher test

Compare crash days together with P(A) × P(B); Fisher suits small counts.

Tail Frequency vs the Normal Distribution



5.48×

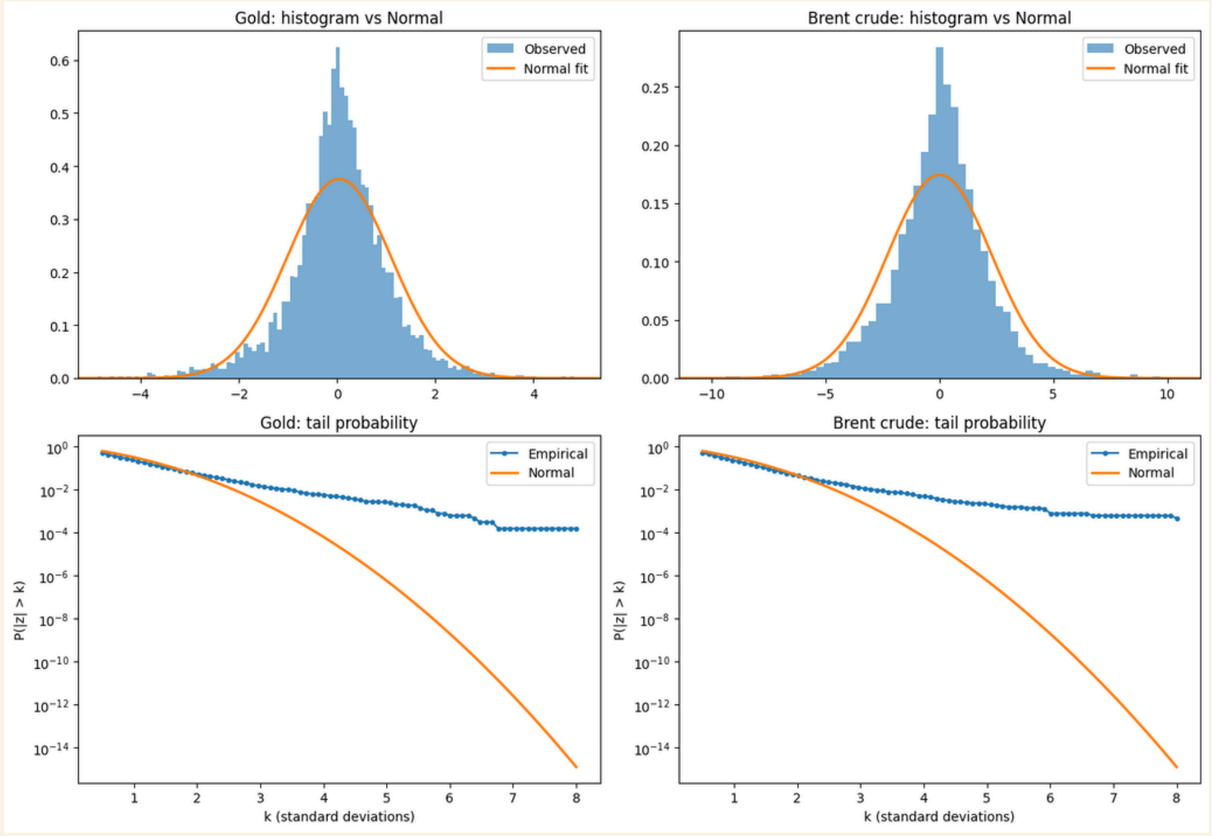
gold: 1.48% observed vs 0.27% Normal

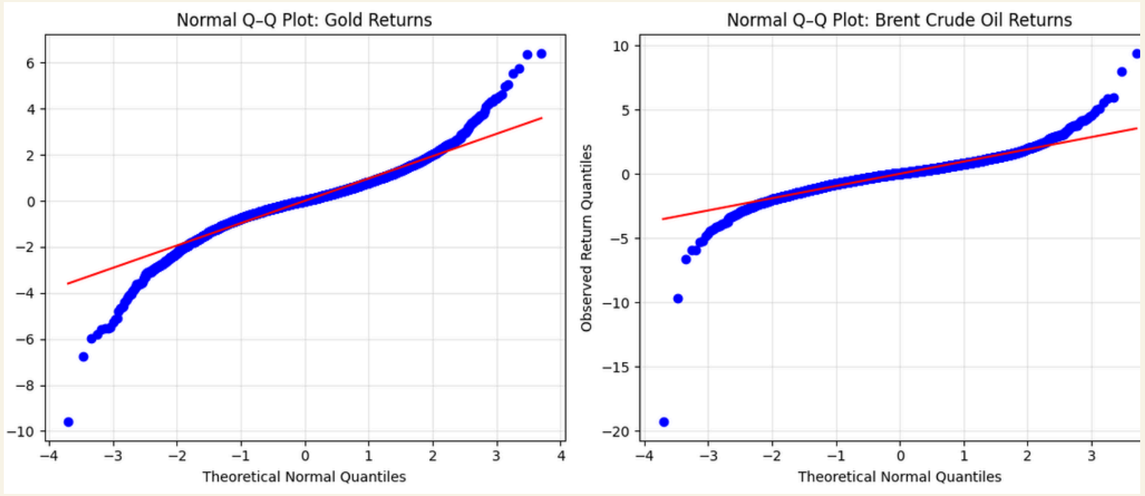
4.45×

crude: 1.20% observed vs 0.27% Normal

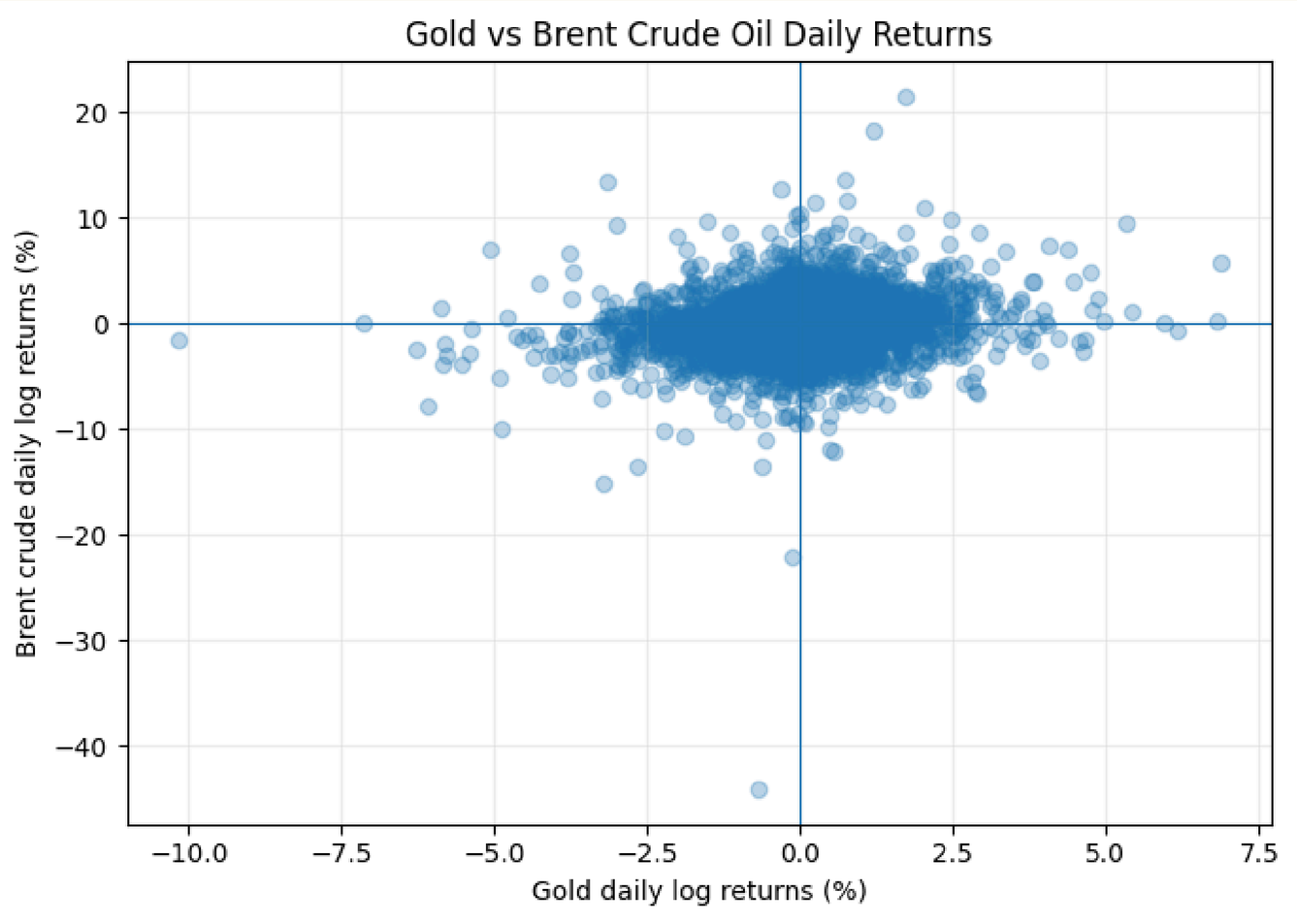
Chebyshev's bound (11.1% at 3 SD) is valid but far too loose to be useful: the real rate is about 1.5%.

Distribution Shape and Q-Q Plots





Correlation and Joint Downside Extremes

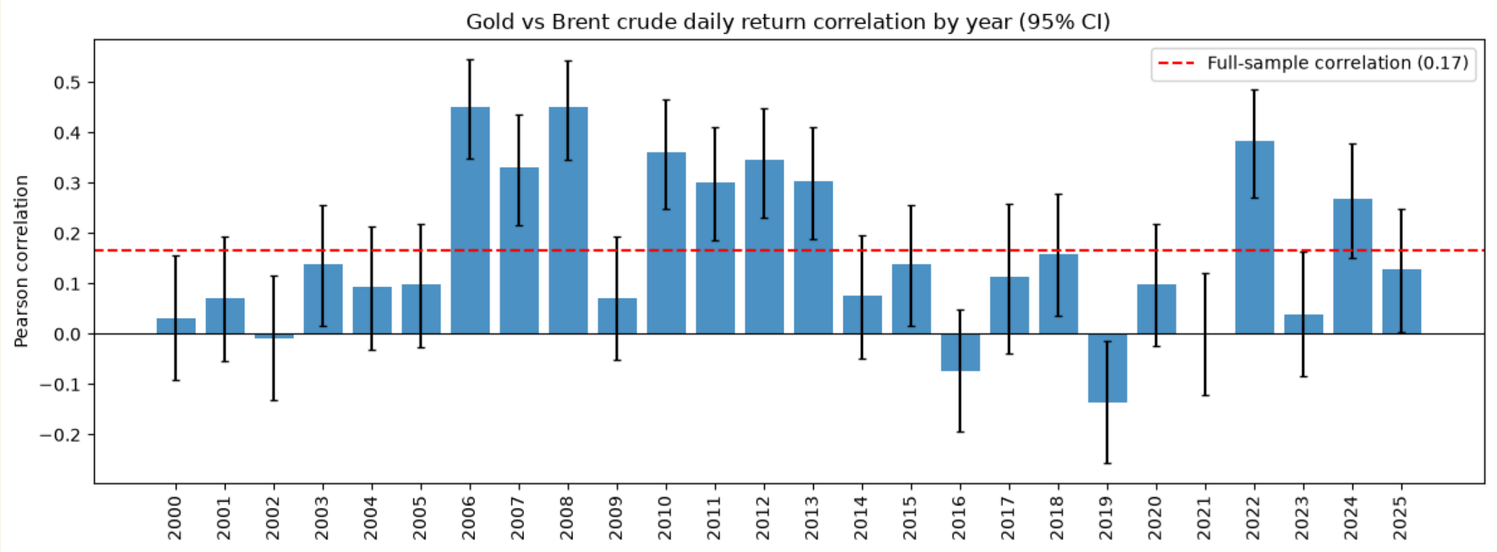


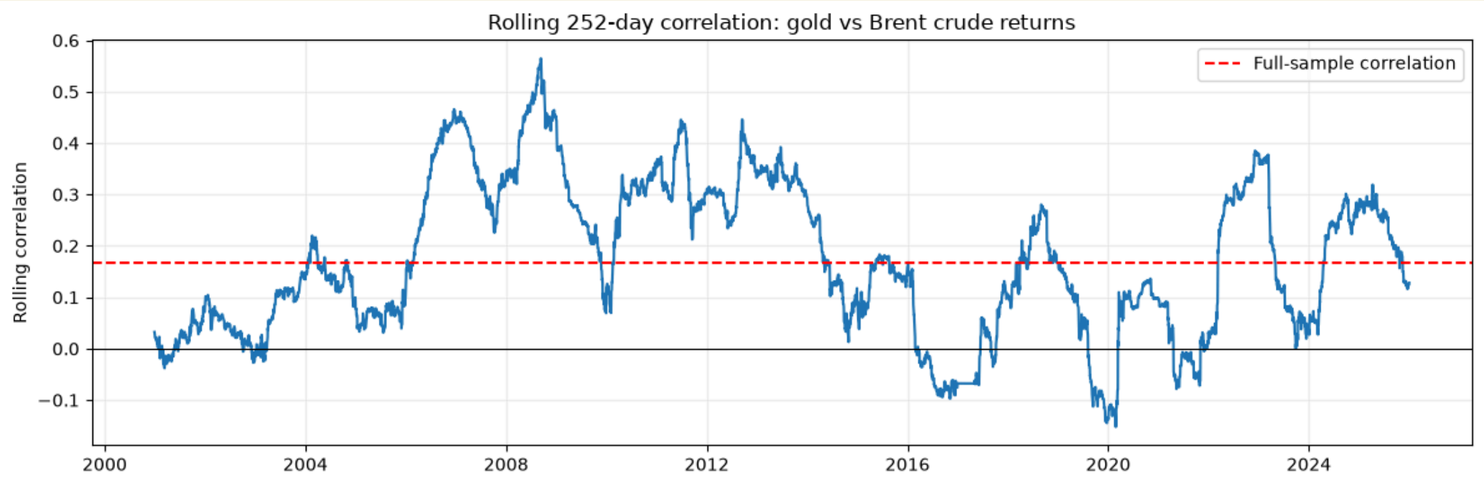
r = 0.17

daily correlation of gold and Brent returns (n = 6,492)

	Crude not extreme	Crude extreme
Gold not extreme	5,883	284
Gold extreme	284	**41**

Gold-Oil Correlation Is Not Stable Over Time

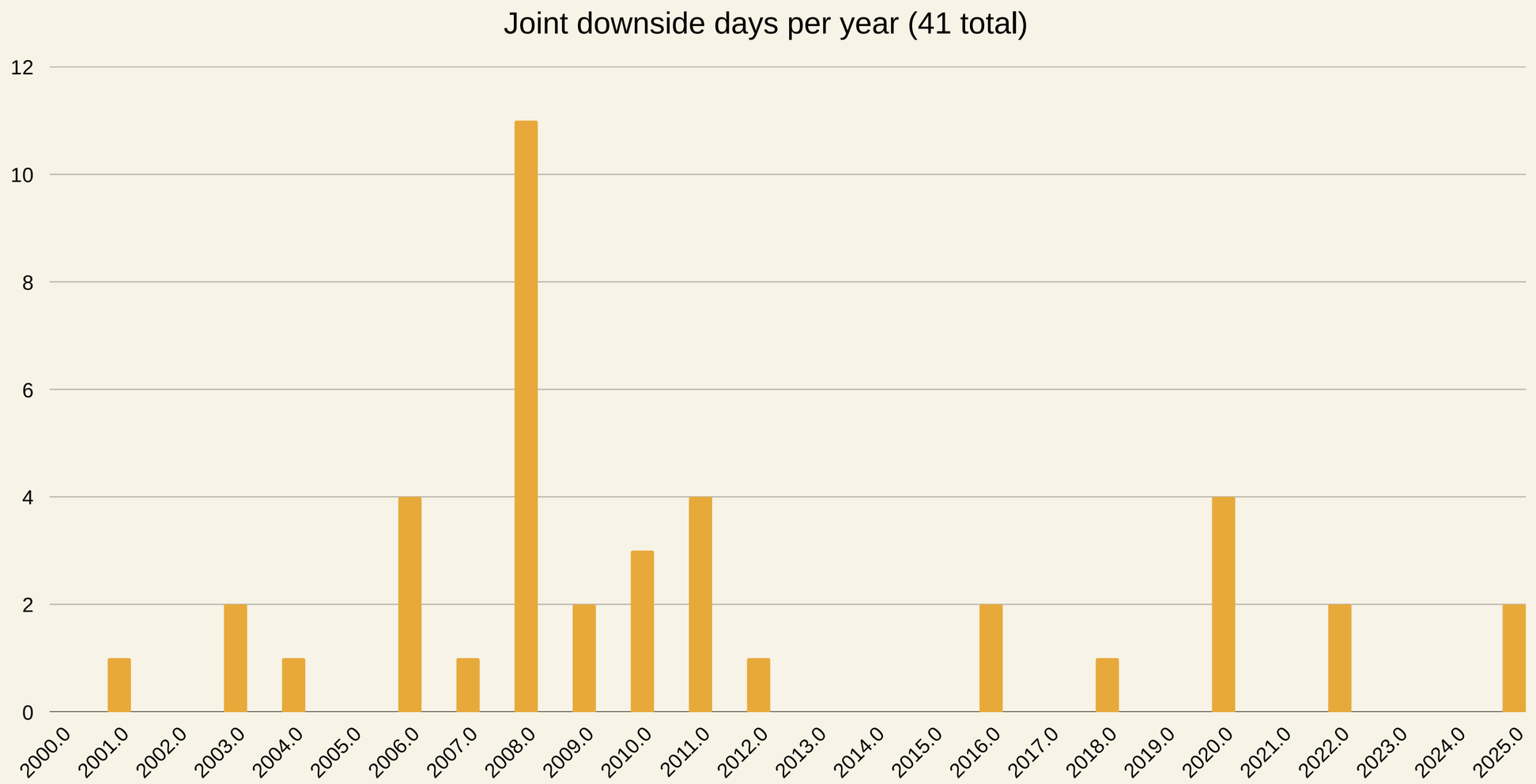




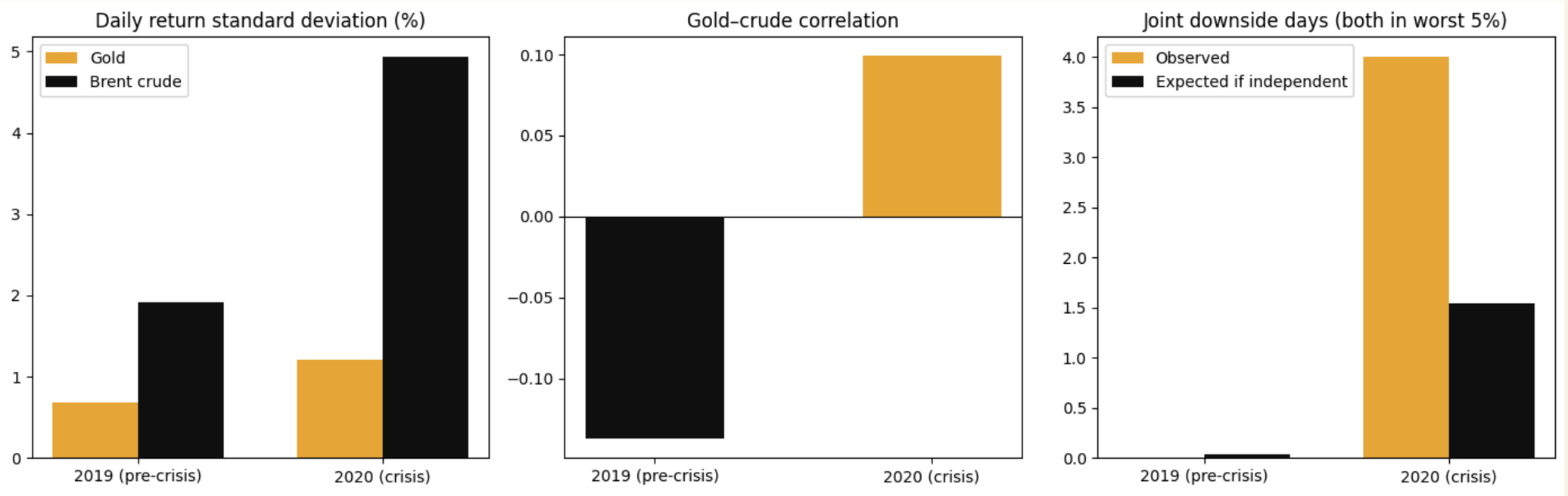
One full-sample number (r = 0.17) hides large year-to-year swings

Strongest years: 2006 (0.45), 2008 (0.45), 2022 (0.38)
Weakest years: 2019 (-0.14), 2016 (-0.07), 2002 (-0.01)
Rolling 252-day correlation: swings from negative to above 0.40
Wide intervals: about 250 days per year, so roughly ±0.12

Joint Downside Days by Year



Pre-crisis vs Crisis Period



2020: 4 joint crash days vs 1.5 expected. Tiny counts, and higher volatility alone adds extremes: suggestive, not proof.

0 → 4

joint crash days: 2019 vs 2020

correlation stays near zero in both years

-0.14 → +0.10

Key Takeaways

1

Bell curves understate tail losses

Consider fat-tailed models (Student's t, extreme value theory) and stress tests for 3-SD days.

2

Low correlation can hide crash risk

Don't judge gold-oil diversification by correlation alone: crashes coincide 2.5× more than chance.

3

Tail risk is not constant

Check sub-periods and crisis windows, not only full-sample averages.

Limitations: what these results can and can't tell us

Days are not independent

Volatility clustering means extreme days bunch together, so Fisher’s p-value is too optimistic.

A few days drive crude

Crude’s kurtosis (27.3) leans on a few 2020 days, led by the -44% drop on 21 Apr 2020.
The full-sample correlation (r = 0.17) hides shifts: -0.14 in 2019, +0.10 in 2020.

Association, not causation

Fisher’s test confirms that tail days co-occur, not that one market causes the other.


Thank
you!